Santiago Avilez Ariza

Senior Full-Stack Software Engineer | Production AI features · TypeScript · Node.js · NestJS · React · Next.js
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Senior full-stack engineer with 5+ years owning production systems end-to-end in TypeScript, React, Next.js and NestJS. Built a multi-tenant SaaS with paying customers from 0 to 1 (Next.js, tRPC, Drizzle/PostgreSQL) and public-facing platforms serving 100,000+ users. Ship production AI features (LLM tool calling, RAG, MCP) and mentor less experienced developers.

EXPERIENCE

Full-Stack Engineer, City of Neuquén - Digital Modernization Office, Neuquén, Argentina, Feb 2025 – Present (full-time)

- Own architecture and technical direction for public-facing platforms (React + NestJS + PostgreSQL) serving 100,000+ users. I define system design, evaluate trade-offs, and drive delivery from problem definition to production with full autonomy.
- Adopted AI-first development workflow using Claude Code and Cursor, spec-driven development (SDD: design, task and spec docs) with orchestrator/sub-agent patterns, reducing implementation cycle time on complex features.
- Led frontend performance refactor, code-splitting, lazy loading, rendering optimizations, reducing load times by **30%** and boosting form completion by **15%**
- Built an MCP server (TypeScript MCP SDK, stdio transport) that indexes the organization's repositories (code, commits, PRs and docs) for RAG, letting developers query the codebase from AI assistants. github.com/santiagoavilez/mcp-repos-rag
- Established automated testing with Jest (~85% coverage on critical modules).
- Mentor junior developers through code reviews, pairing, and architecture guidance, raising code quality across the team.

Stack: TypeScript · React · Next.js · NestJS · Redis · Node.js · PostgreSQL · TypeORM · Docker · Jest · REST APIs

Lead Engineer, Fulbbo | fulbbo.com · Aug 2025 – Present (part-time)

- Built the product 0→1: Next.js + tRPC v11 + Drizzle ORM over PostgreSQL, end-to-end type safety from form field to database.
- Designed the multi-tenant data model (Drizzle schema + migrations) with per-venue isolation enforced by Postgres RLS and org-based roles.
- Paid product: MercadoPago OAuth per venue with idempotent webhook processing covered by Vitest integration tests; Web Push notifications (VAPID).
- Built an AI search layer (OpenAI GPT-4o mini, function calling, Zod) turning natural-language queries into validated filters, cached in Redis.
- LLM-agnostic WhatsApp booking chatbot (Claude Haiku / GPT-4o mini, tool calling): responses validated with Zod and restricted to allow-listed actions, so automation never acts outside defined scope.
- Geographic cache over Google Places (5 km grid), cutting API costs ~90%.

Stack: TypeScript · Next.js · Node.js · tRPC · Drizzle · Supabase (PostgreSQL + Realtime) · Redis · OpenAI · Zod · Vercel · Vitest

Full-Stack Engineer, Zoada Dev Studio (remote) Dec 2020 – Jan 2025 (full-time)

- Built and shipped full-stack applications (React + Next.js + NestJS) for clients in media, e-commerce, education, and startups, owning end-to-end delivery from DB modeling to production deployment.
- Reduced server-side response times by 40% through NestJS query optimization and API response restructuring on a client platform.
- Chose Vertical Slice Architecture over the studio's default Clean Architecture for an e-commerce platform serving 100,000+ users, organizing code by feature to reduce cross-layer coupling and speed up feature delivery.
- Integrated payment gateways and third-party APIs, building RESTful services with error handling, retries, and logging under real production load.

Stack: TypeScript · JavaScript · React · Next.js · NestJS · Node.js · MySQL · PostgreSQL · Jest · REST APIs · Clean Architecture

TECHNICAL SKILLS

Languages: TypeScript · JavaScript · SQL | **Frontend:** React · Next.js · Tailwind CSS · Redux / Zustand
Backend: Node.js · NestJS · tRPC · Express · REST APIs | **Databases:** PostgreSQL · Schema Design · Drizzle · TypeORM · Prisma · MongoDB
Infra & Testing: Docker · AWS (ECS, RDS, S3, EC2) · Vercel · GitHub Actions · CI/CD · Vitest · Jest | **Architecture:** Clean Architecture · Vertical Slice Architecture · SOLID · DDD
Web Accessibility: WCAG 2.1 (keyboard navigation, color contrast, semantic HTML, ARIA attributes)
AI-Assisted Development: LLM integration · RAG · MCP · Structured outputs · Claude Code · Cursor · Spec-Driven Development (SDD) · Multi-agent workflows

EDUCATION

Associate Degree in Web Development — National University of Comahue 2019 – 2022
Systems & Computer Engineering — Universidad de los Andes *(ongoing)* 2023 – Present